MILITARY SPECIFICATION

DEMINERALIZER, WATER, ION EXCHANGE, HAND OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of hand-operated water demineralizer suitable for use in removing dissolved ionizable minerals from water.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Box, Setup.

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MIL-P-116	- Preservation-packaging, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. When not specified, material shall be of a quality which meets the functional requirements of the demineralizer covered by this specification.

3.2 Construction. The demineralizer shall be a hand-operated type consisting of a bottle and a cartridge.

3.2.1 Bottle. Bottle shall have a nominal capacity of 473 milliliters (ml.) (16 ounces) and shall be fabricated of polyethylene. The neck of the bottle shall be so constructed that it holds the cartridge securely.

3.2.2 Cartridge. The cartridge shall be fabricated of rigid plastic and shall contain ion exchange resins and an indicator. Top shall have a protuberance which guides the flow of water from the cartridge. Base shall have a protuberance which fits into the top of the bottle. Sides of cartridge shall be transparent to provide for viewing of indicator. Weight of cartridge shall not exceed 5 ounces (142 grams).

3.2.2.1 Ion exchange resins. Ion exchange resins shall be a mixture of cation and anion resins. An indicator shall be applied to the anion resin before the resins are mixed. Cartridge shall contain sufficient resin to remove 2.21 grams of sodium chloride from a solution of sodium chloride and water.

3.2.2.2 Indicator. Indicator shall be dark blue. Color change of the indicator shall be apparent as a yellow brown band which starts at the base of the cartridge and expands upward as the ion exchange capability of the resins is expended.

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3.2.3 Dimensions. Bottle dimensions shall be: 171 ± 4 millimeters (mm.) high and 70 ± 4 mm. overall outside diameter. Outside diameter of base protuberance on cartridge shall be 16.0 ± .13 mm.

3.3 Performance.

3.3.1 Leakage. When tested as specified in 4.4.1, cartridge and bottle shall remain connected. Water shall pass through the cartridge, and there shall be no leakage from the cartridge, from the bottle or from the area where the bottle and cartridge are connected.

3.3.2 Demineralizing capacity. When tested as specified in 4.4.2, the water which is produced shall contain no particles and no more than 10 parts per million (ppm.) of sodium chloride.

3.4 Marking.

3.4.1 Discard marking. The cartridge shall have a permanent discard mark such as a line or an arrow. Marking shall also indicate that when the yellowish brown band of the indicator (see 3.2.2.2) reaches this mark, cartridge shall be discarded.

3.4.2 Identification marking. The demineralizer or cartridge, as applicable, shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.5 Inspection test date (Military only). The inspection test date shall be 36 months from the date of manufacture.

3.6 Delivery date. Not more than six months shall have elapsed from date of manufacture to date of delivery to the Government.

3.7 Workmanship. Workmanship shall be first class throughout. Demineralizers and cartridges shall be free of defects listed in table III and other defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with table I of MIL-STD-105. Unit of product shall be the individual demineralizer or cartridge, as applicable (see 6.1).

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional and weight examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with table II and MIL-STD-105. Unit of product shall be the demineralizer or replacement cartridge, as applicable. When replacement cartridge is supplied, bottle from a demineralizer which meets the requirements of this specification shall be used for testing purposes. Acceptance number shall be zero.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level
Leakage	3.3.1	4.4.1	S-1
Capacity	3.3.2	4.4.2	S-1

4.3 Examination. Demineralizer or cartridge, as applicable, shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with but shall not be limited to the defects listed in table III.

TABLE III. Classification of defects.

Category	Defect
Major	
101	Component missing.
102	Component not free of tears.
103	Component not free of cracks.
104	Component not free of cuts.
105	Component not free of deformed areas.
106	Discard marking not legible and permanent.
Minor	
201	Identification marking not complete, correct, legible and permanent.

4.3.1 Dimensional and weight examination. The demineralizer or cartridge, as applicable, shall be examined for defects in dimensions and weight. Any dimension or weight not within the tolerances specified herein shall be classified as a defect.

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4.3.2 Packaging inspection. The inspection of the packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Test solution shall be 3.0 grams of sodium chloride in 9.0 liters of distilled or demineralized water.

4.4.1 Leakage. Fill the bottle with test solution. Put cartridge in neck of bottle and squeeze bottle until bottle is empty. Collect the effluent in a suitable container. The same cartridge may be used for capacity test. If used, the volume of effluent produced in the functionability test shall be subtracted from the volume of test solution, 6.6 liters, required for the capacity test.

4.4.2 Demineralizing capacity. Slowly pass 6.6 liters of test solution through the cartridge. Demineralizing capacity testing shall be performed on the last 100 ml. of effluent; particle testing shall apply to all of the effluent. Test solution of 6.6 liters is equivalent to 2.21 grams (35 grains) of sodium chloride (see 3.2.2.1).

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One demineralizer or one box containing two (2) cartridges, as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging.

5.1.1.1.1.1 Demineralizer. Cartridge shall be detached and packaged in accordance with MIL-P-116, method IA-8. Cartridge and bottle shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure of box shall be as specified in the appendix of the applicable box specification.

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5.1.1.1.1.2 Cartridges. Each cartridge shall be packaged in accordance with MIL-P-116, method IA-8. Each unit (2 packaged cartridges) shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676.

5.1.1.1.2 Intermediate packaging. Six demineralizers shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be closed as specified in the appendix of PPP-B-636. Intermediate package is not required for the cartridges specified in 5.1.1.1.1.2.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Seventy two demineralizers or 48 boxes of cartridges shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601 overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Seventy two demineralizers or 48 boxes of cartridges shall be in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. Units shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129. Type II shelf life markings as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Intended use. Demineralizer and replacement cartridge covered by this specification are for use in forward area medical laboratories to remove dissolved ionizable minerals from water used in laboratory procedures.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Items covered by this specification are listed below as shown in the Federal Supply Catalog:

National Stock No.	Item Identification
4610-00-299-9827	DEMINERALIZER, WATER, ION EXCHANGE, Hand-Operated.
4610-00-299-8480	CARTRIDGE, WATER DEMINERALIZER, ION EXCHANGE, Plug-in Type, 4 oz., 2s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 4610-0076

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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